63

A 02000MAT204052101 Pages: 4

Reg No.:_______________ Name:____________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

Course Code: MAT204

Course Name: PROBABILITY, RANDOM PROCESSES AND NUMERICAL METHODS

Max. Marks: 100 Duration: 3 Hours

(Statistical Tables are allowed)

PART A

(Answer all questions; each question carries 3 marks) Marks

1	In a binomial distribution, if the mean is 6, and the variance is 4, find P[X=1].	3
2	Given that f(x) = $\frac{K}{2^x}$ is a probability mass function of a random variable that can take on the values x = 0,1,2,3 and 4, find (i) K and (ii) $P(X \leq 2)$.	3
3	Find the mean and variance for the PDF, f(x) = $\begin{cases} Kx^2 & ,0 < X < 1 \\ 0 & ,elsewhere \end{cases}$	3
4	If random variable X has a uniform distribution in (-3,3), find P ($\lvert X - 2 \rvert < 2$).	3
5	Define stationary random process. Define two types of stationary random process.	3
6	Write down the properties of the power spectral density.	3
7	Write down the Newton's forward and backward difference interpolation formula	3
8	Evaluate $\int_0^1 \frac{1}{1+x^2}dx$ with 4 subintervals by Simpson's rule.	3
9	Write the normal equations for fitting a curve of the form y = a + bx + cx^2 to a given set of pairs of data points.	3
10	Using Euler's method, find y at x = 0.25, given $y^{\prime}$ = 2xy, y (0) =1, h= 0.25.	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	Six dice are thrown 729 times. How many times do you expect at least three dice to show 1 or 2?	6
	b)	Derive the formula for mean and variance of Poisson distribution	8
12	a)	A random variable X takes the values -3, -2, -1,0,1,2,3 such that P(X=0) = P(X>0) = P(X<0) and P(X=-3) = P(X=-2) = P(X=-1) = P(X=1) = P(X=2) = P(X=3). Obtain the probability mass function and distribution function of X.	7

b) The joint probability distribution of X and Y is given by, $f(x, y) = \frac{1}{27}(2x + y)$, $x = 0,1,2$ and $y = 0,1,2$.

(i) Find the marginal distributions of X and Y.

(ii) Are X and Y independent random variables.

Module -2

13 a) Suppose the diameter at breast height (in.) of trees of a certain type is normally distributed with mean 8.8 and standard deviation 2.8.(i) What is the probability that the diameter of a randomly selected tree will be at least 10 in.? (ii) What is the probability that the diameter of a randomly selected tree will exceed 20 in.? (iii) What is the probability that the diameter of a randomly selected tree will be between 5 in. and 10 in.?

b) The amount of time that a surveillance camera will run without having to be reset is a random variable having exponential distribution with mean 50 days. Find the probabilities that such a camera will (a) have to be reset in less than 20 days. (b) not have to be reset in at least 60 days. 7

14 a) The joint density function of 2 continuous random variable X and Y is 7

$$f(x,y)=\begin{cases} cxy \ ; & 0 < x < 4 ,\ 1 < y < 5 \\ 0 \ ; & otherwise \end{cases}$$

(i) Find the value of the constant c.

(ii) Find $P(X \geq 3, Y \leq 2)$

(iii) Find the marginal density of X.

b) The life time of a certain brand of tube light may be considered as a random variable with mean 1200 hours and standard deviation 250 hours. Using Central limit theorem, find the probability that the average life time of 60 lights exceeds 1250. 7

Module -3

15 a) Let $X(t) = A \cos \lambda t + B \sin \lambda t$, where A and B are independent normally distributed random variables $N(0,\sigma^2)$. Show that X(t) is WSS. 7

b) If $X(t) = A \cos(\omega t + \theta)$ Where A and ω are constants and θ is uniformly distributed over $[0, 2\pi]$, find the auto correlation function and Power Spectral Density of the process. 7

16 a) Assume that X(t) is a random process defined as follows: $X(t) = A\cos(2\pi t + \emptyset)$ where A is a zero-mean normal random variable with variance $\sigma_A^2 = 2$ and $\emptyset$ is uniformly distributed random variable over the interval $-\pi \le \varphi \le \pi$. A and φ are statistically independent. Let the random variable Y be defined as $Y = \int_0^1 X(t)dt$. Determine (i) The mean of Y (ii) The variance of Y. 7

b) If the customers arrive at a counter in accordance with Poisson distribution with rate of 2 per minute. Find the probability that the interval between two consecutive arrivals is (i) more than 1 minute (ii) between 1 minute and 2 minutes. 7

Module -4

17 a) Use Newton- Raphson method to find a non- zero solution of $f(x)=2x - \cos x = 0$ 7

b) Using Lagrange's interpolating polynomial estimate y (5) for the following data: 7

x	1	3	4	6
y	-3	0	30	132

18 a) Find the polynomial interpolating the following data, using Newton's backward interpolating formula 7

x	3	4	5	6	7
y	7	11	16	22	29

b) Using Newton's divided difference formula, evaluate y(8) and y(15) from the following data 7

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Module -5

19 a) Solve the following system of equations using Gauss- Seidel iteration method starting with the initial approximation $(0,0,0)^T$ 7

$8x_1 + x_2 + x_3 = 8$

$2x_1 + 4x_2 + x_3 = 4$

$x_1 + 3x_2 + 5x_3 = 5$

b) Fit a straight-line $y = ax + b$ for the following data: 7

X	1	3	4	6	8	9	11	14
Y	1	2	4	4	5	7	8	9

20 a) Solve the following system of equations using Gauss- Jacobi iteration method starting with the initial approximation $(0,0,0)^T$

$$20x_1 + x_2 - 2x_3 = 17$$

$$3x_1 + 20x_2 - x_3 = -18$$

$$2x_1 - 3x_2 + 20x_3 = 25$$

b) Use Runge - Kutta method of fourth order to find y (0.1) from $\frac{dy}{dx} = \sqrt{x+y}$, y (0) =1 taking h = 0.1. 7

***

S_4 - EC, EEE (2019 batch)

KTU July 2021 (2019 scheme)

Course Code : MAT 204

Course Name : Probability, Random Processes and Numerical Methods.

PART - A (Answer all questions. each question carries 3 marks).

1. In a binomial distribution, if the mean is 6, and the variance is 4, find $P[x=1]$.

Ans Given $X \sim b(x, n, p)$, mean $= 6 \Rightarrow np = 6$ —— ①

variance $= 4 \Rightarrow npq = 4$ —— ②

$$\frac{②}{①} \Rightarrow \frac{\cancel{np}q}{\cancel{np}} = \frac{4}{6}$$

$$\Rightarrow q = \frac{4}{6}$$

$$\Rightarrow q = 0.67 \qquad \therefore p = 1 - q$$

$$p = 1 - 0.67$$

$$p = 0.33$$

from ① $\Rightarrow np = 6 \Rightarrow n = \frac{6}{p} \Rightarrow n = \frac{6}{0.33} = 18$.

$\therefore$ binomial p.d.f $\Rightarrow f(x) = n_{C_x}\, p^x\, q^{n-x}$, $x = 0, 1, \ldots n$

$$\Rightarrow f(x) = 18_{C_x} (0.33)^x (0.67)^{18-x}, \; x = 0, 1, \ldots 18$$

$\therefore P[X=x] = f(x) = 18C_x (0.33)^x (0.67)^{18-x}$ —— (3)

$\therefore P[X=1] = 18C_1 (0.33)^1 (0.67)^{17}$

$= 18 \times 0.33 \times (0.67)^{17}$

2 Given that $f(x) = \frac{k}{2^x}$ is a probability mass function of a random variable that can take on the values $x = 0, 1, 2, 3$ and 4, find (i) k (ii) $P[X \le 2]$.

Ans

given $f(x) = \frac{k}{2^x}$, $x = 0, 1, 2, 3, 4.$ —— (1)

x	0	1	2	3	4
$f(x) = \frac{k}{2^x}$	k	$k/2$	$k/4$	$k/8$	$k/16$

we know, $\sum_{\text{all } x} f(x) = 1 \implies k + \frac{k}{2} + \frac{k}{4} + \frac{k}{8} + \frac{k}{16} = 1$

$\implies \frac{31k}{16} = 1$

$\implies k = \frac{16}{31}$

$\therefore f(x) = \frac{16}{31}\left(\frac{1}{2^x}\right)$, $x = 0, 1, 2, 3, 4$ —— (2)

$\therefore P[X \le 2] = P[X = 0, 1, 2]$

$= P[X=0] + P[X=1] + P[X=2]$

$= \frac{16}{31}\left(\frac{1}{2^0}\right) + \frac{16}{31}\left(\frac{1}{2^1}\right) + \frac{16}{31}\left(\frac{1}{2^2}\right) = \frac{28}{31}$ //

3. module II

Find the mean and Variance for the P.d.f, $f(x) = \begin{cases} kx^2, & 0<x<1 \\ 0, & \text{elsewhere.} \end{cases}$

Ans

given $f(x) = \begin{cases} Kx^2, & 0<x<1 \\ 0, & \text{elsewhere} \end{cases}$ (X is continuous rand var)

we know $\Rightarrow \int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\Rightarrow \int_0^1 (Kx^2)\,dx = 1$$

$$\Rightarrow K\left(\frac{x^3}{3}\right)_0^1 = 1$$

$$\Rightarrow K\left[\frac{1}{3}\right] = 1$$

$$\Rightarrow \underline{K=3}$$

$\therefore f(x) = \begin{cases} 3x^2, & 0<x<1 \\ 0, & \text{elsewhere} \end{cases}$

$$\text{mean} = E[X] = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx$$

$$= \int_0^1 x \cdot 3x^2\,dx = 3\int_0^1 x^3\,dx$$

$$= 3\left(\frac{x^4}{4}\right)_0^1 = \frac{3}{4} = \underline{0.75}.$$

$$\text{Variance} = Var(x)$$
$$= E[x^2] - [E[x]]^2$$
$$\text{Variance} = E[x^2] - [0.75]^2 \text{ ——— } (1)$$

$$E[x^2] = \int_{-\infty}^{\infty} x^2 . f(x)\, dx$$
$$= \int_0^1 x^2 . (3x^2)\, dx$$
$$= 3\int_0^1 x^4\, dx$$
$$= 3\left(\frac{x^5}{5}\right)_0^1 = \frac{3}{5}$$

$\therefore (1) \Rightarrow \text{Variance} = Var(x) = \frac{3}{5} - [0.75]^2 = 0.0375$

---

4. (module II) If rand. var. x has a uniform distribution in $(-3, 3)$, find $P[|x-2| < 2]$.

Ans

given $x \sim U(-3, 3)$, $\alpha = -3$, $\beta = 3$.

p.d.f $\Rightarrow f(x) = \frac{1}{\beta - \alpha}$, $\alpha \leq x \leq \beta$. [continuous rand. var.]

$$\Rightarrow f(x) = \begin{cases} \frac{1}{6}, & -3 \leq x \leq 3. \\ 0, & \text{otherwise} \end{cases} \text{ ——— } (1)$$

$$P[|x-2|<2] = P[-2<x-2<2]$$

$$= P[0<x<4]$$

$$= \int_0^4 f(x)\,dx$$

$$= \int_0^4 \left(\frac{1}{6}\right) dx.$$

$$= \frac{1}{6}(x)_0^4 = \frac{4}{6} = 0.666$$

5. Define stationary rand. process. Define two types of stationary random process.

Ans

If certain prob. distribution or statiscal average doesnot depend on time then the rand. process is called Stationary. A rand. Process is called stationary to order n if its n^{th} order prob. density function (pdf) does not change with shift in time origin.

6. Write down the properties of the power spectral density.

Ans

1) The power spectral density and auto correlation function are Fourier transform pairs.

2) Power spectral density (PSD) of a real process $X(t)$ is real, non-negative and even.

3) For a rand. process $X(t)$, the expected power contained in the frequency band $\alpha \leq w \leq \beta$ can be computed as $\frac{1}{2\pi}\int_{\alpha}^{\beta} S_X(w).dw$. In particular, the total expected power is $\frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(w).dw$.

7. Write down the Newton's forward and backward difference interpolation formula.

Ans

Newton's forward difference interpolation formula

$$P_n(x) = y_0 + \frac{u}{1!}(\Delta y_0) + \frac{u(u-1)}{2!}(\Delta^2 y_0) + \frac{u(u-1)(u-2)}{3!}(\Delta^3 y_0) + \cdots$$

$$\cdots + \frac{u(u-1)(u-2)\cdots(u-n+1)}{n!}(\Delta^n y_0).$$

(where $u = \frac{x-x_0}{h}$)

Newton's backward difference interpolation formula.

$$P_n(x) = y_n + \frac{u}{1!}(\nabla y_n) + \frac{u(u+1)}{2!}(\nabla^2 y_n) + \cdots$$

$$\cdots + \frac{u(u+1)(u+2)\cdots(u+n-1)}{n!}(\nabla^n y_n)$$

(where $u = \frac{x-x_n}{h}$)

where $u = \frac{x - x_n}{h}$

8. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ with 4 subintervals by Simpson's rule.

Ans $f(x) = \frac{1}{1+x^2}$, $n = 4$, $a = 0$, $b = 1$, $h = \frac{b-a}{n} = \frac{1}{4}$.

$h = 0.25$

	x_0	x_1	x_2	x_3	x_4
x	0	0.25	0.5	0.75	1
$f(x) = \frac{1}{1+x^2}$	1	0.941	0.8	0.64	0.5
	f_0	f_1	f_2	f_3	f_4

By Simpson's rule,

$$\int_0^1 \frac{dx}{1+x^2} = \frac{h}{3}\left[(f_0 + f_4) + 4(f_1 + f_3) + 2(f_2)\right]$$

$$= 0.7854$$

9. Write the normal equations for fitting a curve of the form $y = a + bx + cx^2$ to a given set of pairs of data points.

Ans $y = a + bx + cx^2$ —— (1)

normal equs are $\Rightarrow \Sigma y = na + b\Sigma x + c\Sigma x^2$

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

10. Using Euler's method, find y at $x=0.25$ given $y'=2xy$, $y(0)=1$, $h=0.25$

Ans given $y'=2xy \Rightarrow f(x,y)=2xy$

$y(0)=1 \Rightarrow x_0=0,\ y_0=1$

$h=0.25$ To find y at $x=0.25$.

ie $y(0.25)$.

ie To find y_1 (I^{st} step)

To find y_1

By Euler's method,

	x_0	x_1
x	0	0.25
y	1	y_1

y_0 I^{st} step

$n=0 \Rightarrow y_1 = y_0 + h f(x_0, y_0)$

$\Rightarrow y_1 = 1 + 0.25(0) = \underline{\underline{1}}$.

$\therefore y$ at $x=0.25$ is $\underline{\underline{1}}$.

Part B

(Answer one full question from each module, each question carries 14 marks)

11. a) Six dice are thrown 729 times. How many times do you expect at least three dice to show 1 or 2.

Ans

6 dice are thrown $(n=6)$

use binomial.

$X \sim b(x, n, p)$

$X \sim b(x, n=6, p=P(\text{occurence of 1 or 2}))$

$X \sim b(x, n=6, p=P[x=1]+P[x=2])$

$X \sim b[x, n=6, p=\frac{1}{6}+\frac{1}{6}] \Rightarrow X \sim b(x, n=6, p=\frac{2}{6})$

Q.2

$X \sim b(x, n=6, p=\frac{1}{3})$ $\therefore q = 1-p = 1-\frac{1}{3} = \frac{2}{3}$

pdf $\Rightarrow f(x) = n_{C_x}\, p^x\, q^{n-x}$, $x = 0, 1, 2 \ldots n$.

$P[X=x] = f(x) = 6_{C_x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}$, $x = 0, 1, 2 \ldots 6$. — (1)

$\therefore$ P[at least 3 dice to show 1 or 2]

" $= P[X \geq 3]$

" $= P[X = 3, 4, 5, 6]$

" $= 1 - P[X = 0, 1, 2]$

" $= 1 - \{P[X=0] + P[X=1] + P[X=2]\}$

" $= 1 - \left\{6_{C_0}\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^6 + 6_{C_1}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^5 + 6_{C_2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^4\right\}$

$P[X \geq 3] = \frac{233}{729} = \underline{\underline{0.319615}}$

$\therefore$ No. of times in 729 throws $= P[X \geq 3] \times 729$

$= 232.99$

$\simeq \underline{\underline{233}}$

b) Derive the formula for mean and variance of Poisson distribution.

Ans

$X \sim f(x, \lambda)$

p.m.f of Poisson distribution is

$$f(x) = \frac{\lambda^x \cdot e^{-\lambda}}{x!}, \quad x = 0, 1, 2 \ldots, \quad \lambda > 0.$$

$$\text{Mean} = E[X] = \sum_{\text{all } x} x \cdot f(x)$$

$$" = \sum_{x=0}^{\infty} x \cdot \left(\frac{\lambda^x e^{-\lambda}}{x!} \right)$$

$$" = 0 + \sum_{x=1}^{\infty} x \cdot \left(\frac{\lambda^x e^{-\lambda}}{x!} \right)$$

$$" = \sum_{x=1}^{\infty} x \cdot \left(\frac{\lambda^x e^{-\lambda}}{x!} \right)$$

$$" = \sum_{x=1}^{\infty} \cancel{x} \cdot \frac{\lambda^x e^{-\lambda}}{\cancel{x}(x-1)!}$$

$$" = \sum_{x=1}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-1)!}$$

$$" = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

$$" = e^{-\lambda} \left[\frac{\lambda}{0!} + \frac{\lambda^2}{1!} + \frac{\lambda^3}{2!} + --- \right]$$

$$" = e^{-\lambda} \cdot \lambda \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + --- \right]$$

$$" = \lambda e^{-\lambda} \left[e^{\lambda} \right]$$

$$\text{Mean} = E[X] = \underline{\underline{\lambda}}$$

$$\text{Variance} = Var(X) = E[X^2] - \{E[X]\}^2$$

$$" = E[X^2] - \{\lambda\}^2$$

$$\text{Variance} = Var(X) = E[X^2] - \lambda^2 \text{ ——— } (1)$$

$$E[X^2] = E[X^2 + X - X]$$

$$'' \quad = E[X(X-1) + X]$$

$$'' \quad = E[X(X-1)] + E[X]$$

$$'' \quad = \sum_{x=0}^{\infty} x(x-1) \cdot f(x) + \lambda$$

$$'' \quad = 0 + 0 + \sum_{x=2}^{\infty} x(x-1) \frac{\lambda^x e^{-\lambda}}{x!} + \lambda$$

$$'' \quad = \sum_{x=2}^{\infty} x(x-1) \frac{\lambda^x e^{-\lambda}}{x!} + \lambda$$

$$'' \quad = \sum_{x=2}^{\infty} \cancel{x}(\cancel{x-1}) \frac{\lambda^x e^{-\lambda}}{\cancel{x}(\cancel{x-1})(x-2)!} + \lambda$$

$$'' \quad = \sum_{x=2}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-2)!} + \lambda$$

$$'' \quad = e^{-\lambda}\left[\frac{\lambda^2}{0!} + \frac{\lambda^3}{1!} + \frac{\lambda^4}{2!} + \cdots\right] + \lambda$$

$$'' \quad = e^{-\lambda} \cdot \lambda^2 \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right] + \lambda$$

$$'' \quad = \lambda^2 e^{-\lambda}\left[e^{\lambda}\right] + \lambda$$

$$E[X^2] = \underline{\underline{\lambda^2 + \lambda}}$$

$\therefore$ ① $\Rightarrow$ Variance $= Var(X) = \lambda^2 + \lambda - (\lambda)^2$

$$= \underline{\underline{\lambda}}$$

12 a) A rand. var. X takes the values $-3,-2,-1,0,1,2,3$ such that $P[X=0]=P[X>0]=P[X<0]$ and $P[X=-3]=P[X=-2]=P[X=-1]=P[X=1]=P[X=2]=P[X=3]$. Obtain the probability mass function and distribution function of X.

Ans

given $X=-3,-2,-1,0,1,2,3$

$$P[X=0]=P[X>0]=P[X<0]$$

$$P[X=0]=P[X=1,2,3]=P[X=-3,-2,-1]$$

$$P[X=0]=P[X=1]+P[X=2]+P[X=3]=P[X=-3]+P[X=-2]+P[X=-1] \quad \text{--- (1)}$$

given $\Rightarrow P[X=-3]=P[X=-2]=P[X=-1]=P[X=1]=P[X=2]=P[X=3]=k \quad \text{--- (2)}$

we know $\Rightarrow \sum f(x)=1$

$$\Rightarrow P[X=-3]+P[X=-2]+P[X=-1]+P[X=0]+P[X=1]+P[X=2]+P[X=3]=1$$

$$\Rightarrow k+k+k+P[X=0]+k+k+k=1$$

$$\Rightarrow 6k+P[X=0]=1$$

$$\Rightarrow P[X=0]=1-6k \quad \text{--- (3)}$$

$\Rightarrow P[X=1]+P[X=2]+P[X=3]=1-6k$ (from (2)) $P[X=0]=P[X=1]+P[X=2]+P[X=3]$

$\Rightarrow k+k+k=1-6k \Rightarrow 3k=1-6k \Rightarrow 9k=1 \Rightarrow k=\frac{1}{9}$

ie

X	-3	-2	-1	0	1	2	3
f(x)	k	k	k	1-6k	k	k	k

Prob. mass. function is

X	-3	-2	-1	0	1	2	3
f(x)	1/9	1/9	1/9	1/3	1/9	1/9	1/9

∴ Distribution function is

X	-3	-2	-1	0	1	2	3
F(x)	1/9	2/9	1/3	2/3	7/9	8/9	1

b) The joint prob. distribution of X and Y is given by $f(x,y) = \frac{1}{27}(2x+y)$; $x = 0, 1, 2$ and $y = 0, 1, 2$. (i) Find the mariginal distributions of X and Y . (ii) Are X and Y independent rand. var.

Ans

given $f(x,y) = \frac{1}{27}(2x+y)$, $x = 0, 1, 2$

$y = 0, 1, 2$.

i) Marginal distributiot X is

$$f_X(x) = \sum_{y=0}^{2} f(x,y)$$

ii) Marginal distribution of Y is

$$f_Y(y) = \sum_{x=0}^{2} f(x,y) .$$

$f(x, y) = \frac{1}{27}(2x+y)$

$x \backslash y$	0	1	2	$f_X(x)$ (marginal density of X)
0	0	1/27	2/27	$3/27 = \underline{\underline{1/9}}$
1	2/27	3/27	4/27	$9/27 = \underline{\underline{1/3}}$
2	4/27	5/27	6/27	$15/27 = \underline{\underline{5/9}}$
marginal density of Y $f_Y(y)$	$\underline{\underline{2/9}}$	$\underline{\underline{1/3}}$	$\underline{\underline{4/9}}$	

(iii) $f_X(x) \cdot f_Y(y) = f(x, y)$ then, X and Y are independent

$f_X(0) \cdot f_Y(0) = 0 = f(0,0)$.

$f_X(0) \cdot f_Y(1) = \frac{1}{9} \times \frac{1}{3} = \frac{1}{27} = f(0,1)$

$\therefore$ All combination are satisfied

$\therefore$ X and Y are independent

13 a) Suppose the diameter at breast height (in) of trees of a certain type is normally distributed with mean 8.8 and standard deviation 2.8. (i) What is the prob. that the diameter of a randomly selected tree will be at least 10 in. (ii) what is the prob. that the diameter of a randomly selected tree will exceed 20 in. (iii)

what is the prob. that the diameter of a randomly selected tree will be between 5 in and 10 in.

Ans

$X \sim N(\mu, \sigma^2) \xrightarrow[\text{to}]{\text{convert}} Z \sim N(0,1)$. given mean = 8.8 and std. deviation = 2.8

$$Z = \frac{X-\mu}{\sigma}$$

$$Z = \frac{X-8.8}{2.8}$$

i) P[diameter of a randomly selected tree will be at least 10 in] $= P[X \geq 10]$

$$= P\left[\frac{X-\mu}{\sigma} \geq \frac{10-\mu}{\sigma}\right]$$

$$= P\left[Z \geq \frac{10-8.8}{2.8}\right]$$

$$= 1 - P\left[Z < \frac{10-8.8}{2.8}\right]$$

$$= 1 - P[Z < 0.43] = 1 - F(0.43)$$

$$= \underline{0.336}$$

(ii) P[diameter of a randomly selected tree will be exceeds 20 in] $= P[X > 20] = P\left[Z > \frac{20-8.8}{2.8}\right]$

$= \text{approximately } \underline{0}$

(iii) P[diameter of a randomly selected tree will be between 5 in and 10 in] $= P[5 < X < 10] = P\left[\frac{5-8.8}{2.8} < Z < \frac{10-8.8}{2.8}\right]$

$$= F\left[\frac{5-8.8}{2.8}\right] - F\left[\frac{10-8.8}{2.8}\right] = \underline{0.5795}$$

b) The amount of time that a surveillance camera will run without having to be reset is a random variable having exponential distribution with mean 50 days. Find the prob. that such a camera will (a) have to be reset in less than 20 days. (b) not have to be reset in atleast 60 days.

Ans exponential distribution
mean = 50

ie $\beta = 50$ $\quad\quad \beta = \frac{1}{\lambda}$

$$f(x) = \frac{1}{50} e^{-x/50}, \quad x > 0 \quad \beta > 0$$

(a) $P[\text{reset in less than } 20] = P[X \leq 20] = \int_{-\infty}^{20} f(x)\,dx$

$$= \int_0^{20} \frac{1}{50} e^{-x/50}\,dx$$

$$= \underline{\underline{0.3297}}$$

(b) $P[\text{not have to be reset in atleast } 60] = P[X \geq 60] = \int_{60}^{\infty} f(x)\,dx.$

$$= \int_{60}^{\infty} \frac{1}{50} e^{-x/50}\,dx$$

$$= \underline{\underline{0.3012}}.$$

a) The joint density function of 2 continuous rand. var. X and Y is $f(x,y)=\begin{cases} cxy, & 0<x<4, \\ & 1<y<5 \\ 0, & \text{otherwise} \end{cases}$

(i) Find the value of the constant c. (ii) Find $P[X\geq 3, Y\leq 2]$ (iii) Find marginal density of X.

Ans

(i) $\int_0^4 \int_1^5 cxy \cdot dy\,dx = 1 \Rightarrow c\left(\frac{x^2}{2}\right)_0^4 \left(\frac{y^2}{2}\right)_1^5 = 1$

$\Rightarrow c = \frac{1}{96}$

$\therefore f(x,y) = \begin{cases} \frac{xy}{96}, & 0<x<4, \ 1<y<5 \\ 0, & \text{otherwise} \end{cases}$

(ii) $P[X\geq 3, Y\leq 2] = \int_3^4 \int_1^2 \frac{xy}{96}\, dy\, dx$

$= \frac{1}{96}\left(\frac{x^2}{2}\right)_3^4 \left(\frac{y^2}{2}\right)_1^2$

$= \frac{1}{96}\left(\frac{16}{2}-\frac{9}{2}\right)\left(\frac{4}{2}-\frac{1}{2}\right)$

$= \frac{1}{96} \times \frac{7}{2} \times \frac{3}{2} = \frac{7}{128}$

(iii) marginal density of $X = f_X(x)$

$= \int_y f(x,y)\,dy$

$= \int_1^5 \frac{xy}{96}\,dy = \frac{x}{96}\left(\frac{y^2}{2}\right)_1^5$

$= \frac{x}{96}\left(\frac{25}{2}-\frac{1}{2}\right) = \frac{x}{8}, \quad 1<y<5$

b) The life time of a certain brand of tube light may be considered as a random variable with mean 1200 hours and standard deviation 250 hours. Using central limit theorem, find the prob. that the average life time of 60 lights exceeds 1250.

Ans.

given $E[X_n] = 12000 = (\mu)$ $\quad n = 60$

$$\text{std. deviation} = (\sigma)_{(X_n)} = 250$$

$$\therefore \text{variance} = Var(X_n) = (250)^2 = (\sigma)^2$$

average life time $\bar{X} \sim N\left(\text{mean} = \mu,\ \text{variance} = \frac{\sigma^2}{n}\right)$

$$Z = \frac{\bar{X} - \mu}{\sigma}$$

$\therefore$ mean $= \mu = \underline{12000}$.

$$\text{Variance} = \frac{(250)^2}{(60)} = \sigma^2 \implies \sigma = \sqrt{\frac{(250)^2}{60}} = \frac{250}{\sqrt{60}}$$

$$\therefore Z = \frac{\bar{X} - 12000}{(250/\sqrt{60})}$$

$\therefore P[\text{average life time of 60 lights exceeds 1250}]$

$$= P[\bar{X} \geq 1250] = P\left[Z \geq \frac{1250 - 12000}{250/\sqrt{60}}\right]$$

$$= P[Z \geq 1.5] = 1 - P[Z < 1.5] = \underline{\underline{0.0606}}$$

Module - 3

5. a) Let $X(t) = A\cos\alpha t + B\sin\alpha t$, where A and B are independent normally distributed rand. variables $N(0, \sigma^2)$. Show that $X(t)$ is WSS.

Ans.

given $X(t) = A\cos\alpha t + B\sin\alpha t$

where A and B are independent normally distributed rand. var. $N(0, \sigma^2)$

ie $E[A] = E[B] = 0$ and

$Var[A] = Var[B] = \sigma^2$.

$\therefore E[A^2] = E[B^2] = \sigma^2$.

$\therefore E[X(t)] = E[A\cos\alpha t + B\sin\alpha t]$

$\quad = E[A]\cos\alpha t + E[B]\sin\alpha t$.

mean of $X(t) = \underline{\underline{0}}$,

Auto correlation $= R_{XX}(t_1, t_2)$

$= E[X(t_1) \cdot X(t_2)]$

$= E[(A\cos\alpha t_1 + B\sin\alpha t_1)(A\cos\alpha t_2 + B\sin\alpha t_2)]$

$= E[A^2\cos\alpha t_1 \cdot \cos\alpha t_2 + AB\cos\alpha t_1 \cdot \sin\alpha t_2 + AB\sin\alpha t_1 \cdot \cos\alpha t_2 + B^2\sin\alpha t_1 \sin\alpha t_2]$

$= E[A^2] \cdot E[\cos\alpha t_1 \cdot \cos\alpha t_2] + E[AB]\, E[\cos\alpha t_1 \cdot \sin\alpha t_2] +$

$E[AB] \cdot E[\sin\alpha t_1 \cdot \cos\alpha t_2] + E[B^2] \cdot E[\sin\alpha t_1 \cdot \sin\alpha t_2]$

$= E[A^2] \cdot \cos\alpha t_1 \cdot \cos\alpha t_2 + 0 + 0 + E[B^2] \cdot \sin\alpha t_1 \cdot \sin\alpha t_2$

$= E[A^2] \cdot \cos\alpha t_1 \cdot \cos\alpha t_2 + E[B^2] \cdot \sin\alpha t_1 \cdot \sin\alpha t_2$

$= \sigma^2 \cos\alpha t_1 \cdot \cos\alpha t_2 + \sigma^2 \sin\alpha t_1 \cdot \sin\alpha t_2$. (α is constant) (not a rand. var)

$= \sigma^2 [\cos(\alpha t_1 - \alpha t_2)]$

$= \sigma^2 \cos\alpha(t_1 - t_2)$ interms of time difference

$\therefore$ It is WSS

---

b) If $X(t) = A\cos(\omega t + \theta)$, where A and ω are constants and θ is uniformly distributed over $[0, 2\pi]$, find the auto correlation function and power spectral density of the process.

Ans given $X(t) = A\cos(\omega t + \theta)$.

$X \sim U(\alpha, \beta)$

$f(x) = \frac{1}{\beta - \alpha}, \alpha < x < \beta$

$A, \omega \Rightarrow$ constant

$\theta \sim U(0, 2\pi)$ is rand. var $\Rightarrow f(\theta) = \frac{1}{2\pi}, 0 \le \theta \le 2\pi$

Autocorrelation $R_{XX}(t_1, t_2) = E[X(t_1) \cdot X(t_2)]$

$$= E[A\cos(\omega t_1 + \theta) \cdot A\cos(\omega t_2 + \theta)]$$

$$= E[A^2 \cdot \cos(\omega t_1 + \theta) \cdot \cos(\omega t_2 + \theta)]$$

$$= E\left[A^2 \cdot \frac{\cos[\omega(t_1+t_2)+2\theta] + \cos\omega(t_1-t_2)}{2}\right]$$

$$= A^2 \cdot E\left[\frac{\cos(\omega(t_1+t_2)+2\theta) + \cos\omega(t_1-t_2)}{2}\right]$$

$$= \frac{A^2}{2} \cdot E\left[\cos[\omega(t_1+t_2)+2\theta] + \cos\omega(t_1-t_2)\right]$$

$$= \frac{A^2}{2} \int_0^{2\pi} \left[\cos[\omega(t_1+t_2)+2\theta] + \cos\omega(t_1-t_2)\right] \cdot f(\theta) \cdot d\theta$$

$$= \frac{A^2}{2} \int_0^{2\pi} \left[\cos[\omega(t_1+t_2)+2\theta] + \cos\omega(t_1-t_2)\right] \times \frac{1}{2\pi} \, d\theta$$

$$= \frac{A^2}{4\pi} \int_0^{2\pi} \left[\cos[\omega(t_1+t_2)+2\theta] + \cos(\omega(t_1-t_2)\right] d\theta$$

$$= \frac{A^2}{4\pi} \left[\frac{\cos[\omega(t_1+t_2)+2\theta]}{2}\right]_{\theta=0}^{2\pi} + \cos\omega(t_1-t_2)[\theta]_{\theta=0}^{2\pi}$$

$$= \frac{A^2}{2} \cos\omega(t_1-t_2)$$

$$= \frac{A^2}{2} \cos\omega\tau \qquad \tau = t_1 - t_2.$$

$\therefore$ Power spectral density of the process $= S_X(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) \cdot e^{-i\omega\tau} \cdot d\tau$.

$= \text{Fourier transform of} \left[R_{XX}(t_1, t_2)\right]$

$= F\left[R_{XX}(\tau)\right]$

$= F\left[\frac{A^2}{2} \cos \omega \tau\right]$

$= \frac{A^2}{2} F\left[\cos \omega \tau\right]$

$= \frac{A^2}{2} F\left[\cos A \tau\right]$

$= \frac{A^2}{2}\left[\pi\, \delta(\omega - A) + \delta(\omega + A)\right]$

$= \frac{A^2}{2}\left[\pi\, \delta(\omega - \omega) + \delta(\omega + \omega)\right] = \frac{A^2}{2}\left[\pi\, \delta(2\omega) + \delta(0)\right]$

Fourier transform of $\cos A \tau$
$= \pi\left[\delta(\omega - A) + \delta(\omega + A)\right]$

here $\omega =$ constant
Replace $\omega = A$

16. a) Assume that $X(t)$ is a rand. process defined as follows $X(t) = A \cos(2\pi t + \phi)$ where A is zero-mean normal rand. var. with variance $\sigma_A^2 = 2$ and ϕ is uniformly distributed rand. var. over the interval $-\pi \leq \phi \leq \pi$. A and ϕ are statistically independent. Let the rand. var. Y be

defined as $Y = \int_0^1 X(t) \cdot dt$. Determine (i) The mean of Y (ii) The variance of Y.

<u>Ans</u> $X(t) = A\cos(2\pi t + \phi)$

$A \sim N(\text{mean} = 0, \text{variance} = 2)$

$\phi \sim U(-\pi, \pi)$

A and ϕ are statistically independent.

mean of $Y = E[Y]$

" $= E\left[\int_0^1 X(t)\, dt\right]$

" $= \int_0^1 E[X(t)] \cdot dt$

" $= \int_0^1 (0)\, dt$

$E[Y] = \underline{\underline{0}}$

$E[X(t)] = E[A\cos(2\pi t + \phi)]$

$= E[A] \cdot E[\cos(2\pi t + \phi)]$

$= \underline{\underline{0}}$.

variance of $X(t) = \sigma^2_{X(t)} = \underline{\underline{1}}$.

~~$E[Y]$~~ = variance of $Y = Var(Y) = E[Y^2] - (E[Y])^2$

$\Rightarrow Var(Y) = E[Y^2] - (0)^2$

$\Rightarrow Var(Y) = E[Y^2]$

$\Rightarrow Var(Y) = \underline{\underline{0}}$.

b) If the customers arrive at a counter in accordance with Poisson distribution with rate of 2 per minute. Find the prob. that the interval between two consecutive arrivals

(i) more than 1 minute (ii) between 1 minute and 2 minute.

Ans

question about time.

exponential. $T \sim exp(\lambda)$, $f(t) = \lambda e^{-\lambda t}$, $\lambda = 2$

i) $P[T > 1] = 1 - P[T \leq 1] = \underline{0.135}$

ii) $P[1 < T < 2] = \underline{0.117}$

Module-4.

17. a) Use Newton-Raphson method to find a non-zero solution $f(x) = 2x - \cos x = 0$

Ans $f(x) = 2x - \cos x$ $\quad f'(x) = 2 + \sin x$.

$f(0) = -ve$

$f(1) = 1.4596\ (+ve)$

$x_0 = \frac{a+b}{2} = \frac{1}{2} = 0.5$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$x_1 = 0.5$

$x_2 = 0.45063$ $\quad x_3 = 0.45018$ $\quad \underline{x_4 = 0.4502}$.

b) Using Lagrange's interpolating polynomial estimate $y(5)$ for the following data

x	1	3	4	6
y	-3	0	30	132

Ans $y = L_0(x)\, y_0 + L_1(x)\, y_1 + L_2(x)\, y_2 + L_3(x) \cdot y_3$

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$\therefore\ y(5) = 75$

18 a) Find the polynomial interpolating the following data using Newtons backward interpolating formula.

x	3	4	5	6	7
y	7	11	16	22	29

Ans $P_n(x) = y_n + \frac{u}{1!} \nabla y_n + \frac{u(u+1)}{2!} \nabla^2 y_n + \cdots$

$$u = \frac{x - x_n}{h} = x - 7.$$

$$P_4(x) = y_n + \frac{u}{1!}(\nabla y_n) + \frac{u(u+1)}{2!}(\nabla^2(y_n)) + \frac{u(u+1)(u+2)}{3!}(\nabla^3 y_n)$$

$$+ \frac{u(u+1)(u+2)(u+3)}{4!}(\nabla^4 y_n)$$

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
3	7				
		4			
			1		
4	11			0	
		5			
					0
5	16		1		
				0	
		6			
6	22		1		
		7			
7	29				

$$\therefore P_n(x) = \frac{1}{2}(x^2+x+2)$$

b) Using newton's divided difference formula, evaluate $y(8)$ and $y(15)$ from the following data

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Ans

x	y	I divided difference	2nd	3rd	4th	5th
4	48					
		52				
			15			
5	100			1		
		97			0	
7	294		21			0
		202		1		
					0	
10	900		27			
		310		1		
11	1210		33			
		409				
13	2028					

$y(8) = 448$, $y(15) = 3150$

a) Solve the following system of equations using Gauss-Seidel iteration method starting with the initial approximation $(0,0,0)^T$.

$8x_1 + x_2 + x_3 = 8$, $2x_1 + 4x_2 + x_3 = 4$, $x_1 + 3x_2 + 5x_3 = 5$

Ans

$$x_1 = \frac{1}{8}[8 - x_2 - x_3]$$

$$x_2 = \frac{1}{4}[4 - 2x_1 - x_3]$$

$$x_3 = \frac{1}{5}[5 - x_1 - 3x_2]$$

1^{st} iteration $\Rightarrow x_1 = 1$, $x_2 = 0.5$, $x_3 = 0.5$

2^{nd} iteration $\Rightarrow x_1 = 0.875$, $x_2 = 0.4375$, $x_3 = 0.5625$

3^{rd} iteration $\Rightarrow x_1 = 0.875$ $x_2 = 0.4219$, $x_3 = 0.572$

4^{th} iteration $\Rightarrow x_1 = 0.872$ $x_2 = 0.419$, $x_3 = 0.573$

5^{th} iteration $\Rightarrow x_1 = 0.876$ $x_2 = 0.419$ $x_3 = 0.573$

b) Fit a st. line $y = ax + b$ for the following data.

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

Ans $y = ax + b$

normal equs $\Rightarrow \Sigma y = a\Sigma x + nb$

$$\Sigma xy = a\Sigma x^2 + b\Sigma x.$$

$a \approx 0.64$, $b \approx 0.55$ (from the table and solve)

$\therefore y = 0.64x + 0.55$

20. a) Solve the following system of equations using Gauss-Jacobi iteration method starting with the initial approximation $(0,0,0)^T$

$$20x_1 + x_2 - 2x_3 = 17$$
$$3x_1 + 20x_2 - x_3 = -18$$
$$2x_1 - 3x_2 + 20x_3 = 25$$

Ans $x_1 = \frac{1}{20}[17 - x_2 + 2x_3]$

$x_2 = \frac{1}{20}[-18 - 3x_1 + x_3]$

$x_3 = \frac{1}{20}[25 - 2x_1 + 3x_2]$

I^{st} iteration $\Rightarrow x_1 = 0.85,\ x_2 = -0.9,\ x_3 = 1.25$

2^{nd} $\Rightarrow x_1 = 1.02,\ x_2 = -0.965,\ x_3 = 1.03$

3^{rd} $\Rightarrow x_1 = 1.00125,\ x_2 = -1.0015,\ x_3 = 1.00325$

4^{th} $\Rightarrow x_1 = 1.004,\ x_2 = -1,\ x_3 = 0.9965$

5^{th} $\Rightarrow x_1 = 1,\ x_2 = -1,\ x_3 = 1.$

b) Use R-K method of 4^{th} order to find $y(0.1)$ from $\frac{dy}{dx} = \sqrt{x+y}$, $y(0) = 1$ taking $h = 0.1$

Ans

$f(x, y) = \sqrt{x+y}$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$

$x_0 = 0$, $x_1 = 0.1$, $x_2 = 0.2$

$K_1 = 0.1$, $K_2 = 0.10488$, $K_3 = 0.10499$,

$\therefore$ $y_1 = 1.1049$